

MP-03 HVAC Filter and Airflow Service Manual
Category: Maintenance Procedures
Synthetic sample document for public portfolio review.

DOCUMENT CONTROL

Document ID: MP-03-HVAC-RTU7
Revision: Rev 1.0
Owner: Facilities Reliability Team
Applies To: North Campus Distribution Operations
Review Cycle: Every 12 months or after equipment redesign

PURPOSE

This manual defines the quarterly service routine for the fictional RTU-7 rooftop HVAC unit serving office and staging areas. It covers filter replacement, condensate inspection, belt condition, airflow verification, and restart documentation.

The procedure is intended for trained maintenance technicians and shift leads. Production operators may assist with observation and restart confirmation, but they do not perform isolated service work unless the task card explicitly allows it.

REQUIRED TOOLS AND PARTS

- Roof access permit, multimeter, differential pressure gauge, spare filters, belt tension tool, flashlight, and hand brush.
- Approved replacement filters with the exact MERV rating listed on the asset card.
- Portable radio or phone for contact with the building coordinator during shutdown and restart.
- Weather-appropriate fall protection and non-slip footwear for roof access.

HAZARDS AND CONTROLS

- Do not access the roof during lightning, freezing precipitation, or sustained high wind conditions.
- Use the local disconnect to isolate the unit before opening service panels.
- Take care around hot surfaces near compressor lines and sharp cabinet edges inside filter and blower compartments.
- If standing water is present near electrical sections, stop work and escalate before energizing the unit.

PREPARATION

- Coordinate a service window with the building coordinator so occupants understand the temporary shutdown.
- Isolate the RTU, verify fan stop, and inspect exterior panel condition, seals, and visible fasteners before opening the cabinet.
- Remove used filters, compare size and rating to the asset card, and install matched

replacements with airflow arrows oriented correctly.

- Inspect blower belt wear, pulley alignment, condensate drain condition, and coil face cleanliness. Brush away loose debris without bending fins.
- Measure differential pressure across the new filters after restart and compare it with the acceptable post-service band on the asset card.
- Check supply and return temperature difference after the unit stabilizes for ten minutes.
- Confirm all access panels are secured and that no tools or packaging remain inside the cabinet before final restart.

PROCEDURE

- Installed filters match size and MERV requirement with no bypass gaps.
- Condensate path is clear and no standing water remains in the drain pan.
- Belt tension is within the approved range and no abnormal vibration or squeal is present after restart.
- Measured temperature split and filter differential pressure are within the unit baseline or have a documented reason for variance.

ACCEPTANCE CRITERIA

- Record filter size, quantity, and lot placeholder used during service.
- Capture ambient outdoor temperature and supply-return delta at the time of final readings.
- List any deferred roof or cabinet repairs separately from the routine filter task.
- Attach photos only if the project later introduces media support; for Phase 1 and Phase 2 planning, note conditions in text.

RECORDS AND CLOSEOUT

- A realistic retrieval question for this manual would ask whether the temperature split is taken immediately after restart or after a stabilization period. The answer appears in the procedure steps and not in the acceptance criteria, which helps test precise grounding.
- Another useful scenario involves the wrong filter rating arriving from stores. The manual implies the unit should not be closed out with a mismatched filter, even if airflow appears normal.

EXAMPLES AND NOTES

APPENDIX 1: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 2: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 3: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 4: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 5: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 6: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 7: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 8: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.